

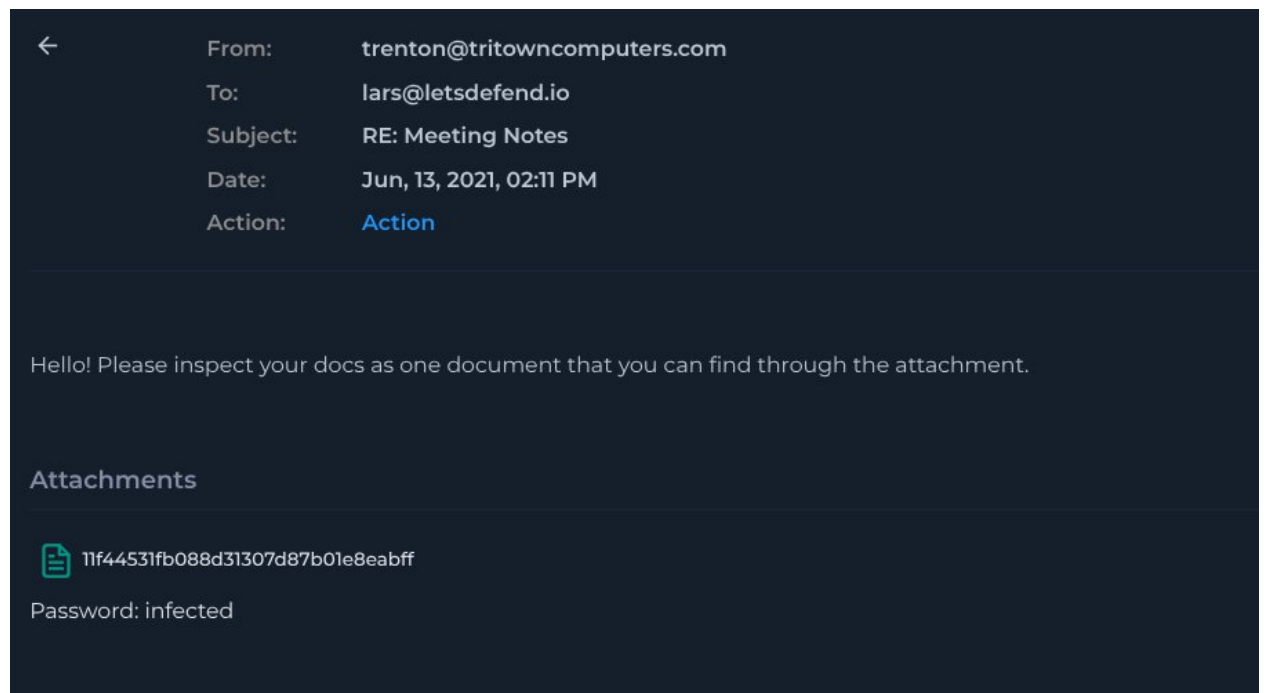
Event Information:

☆ This alert was generated from a real phishing attack.

EventID :
93
Event Time :
Jun, 13, 2021, 02:13 PM
Rule :
SOC146 - Phishing Mail Detected - Excel 4.0 Macros
Level :
Security Analyst
SMTP Address :
24.213.228.54
Source Address :
trenton@tritowncomputers.com
Destination Address :
lars@letsdefend.io
E-mail Subject :
RE: Meeting Notes
Device Action :
Allowed

Analysis Steps:

Email logs were searched to find the malicious email.



The file was downloaded in order to obtain hashes.

File	MD5Hash
iroto.dll	e03bde4862d4d93ac2ceed85abf50b18
iroto1.dll	8e6fbefcbac2a1967941fa692c82c3ca
research-1646684671.xls	b775cd8be83696ca37b2fe00bcb40574

None of these hashes are in the LetsDefend CTI database.

iroto.dll is listed as malicious on VirusTotal by 18/62 Vendors of the type Trojan and family labels abtrojan, dsk and qshell as of 4 days ago.

18 / 62
Community Score

18/62 security vendors flagged this file as malicious

05b9e3a987aec9ba7adb0ee947d96516a213d663cc5a71c78af146a946
iroto.dll
pedi checks user input detect debug environment overlay

Size
434.56 KB

Last Analysis Date
4 days ago

Reanalyze Similar More

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 11

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.abtrojan/dsk Threat categories trojan Family labels abtrojan dsk qshell

Security vendors' analysis Do you want to automate checks?

Arctic Wolf	Unsafe	Avira (no cloud)	TR/W32.Evo
Bkav Pro	W32/Malware-293CB9F7	CTX	DLL.Trojan.artemis
Fortinet	W32/Qshell.VHO/tr	Google	Detected
Lionic	Trojan.Win32.Generic.A/c	McAfee Scanner	TIE058B0E9AF187
Palo Alto Networks	Generic.mdl	Sangfor Engine Zero	Trojan.Win32.Evo.Vc2z
SecureAge	Malicious	Skyhigh (SWG)	Artemis

iroto1.dll is listed as malicious on VirusTotal by 21/68 Vendors of the type Trojan and family labels usblfp26, abtrojan and dslu as of 9 days ago.

21 / 68
Community Score

21/68 security vendors flagged this file as malicious

e05c717b43f7e204f315eb8c298f715791385516335acd8f2ec9e26c3e960b
iroto1.dll
pedi detect debug environment overlay checks user input

Size
434.52 KB

Last Analysis Date
9 days ago

Reanalyze Similar More

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 12

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.usblfp26/abtrojan Threat categories trojan Family labels usblfp26 abtrojan dslu

Security vendors' analysis Do you want to automate checks?

Avast	Win32:Evo-gen [Trj]	AVG	Win32:Evo-gen [Trj]
Avira (no cloud)	TR/W32.Evo	Bkav Pro	W32/Malware.E093F271
CTX	DLL.Trojan.artemis	Cyren	Malicious (score: 99)
Fortinet	W32/Qshell.VHO/tr	Google	Detected
Lionic	Trojan.Win32.Generic.A/c	McAfee Scanner	TIE05C717B43F7
Palo Alto Networks	Generic.mdl	Sangfor Engine Zero	Trojan.Win32.Evo.V7mo

What is likely the main file that the attacker hopes that the user will access research-1646684671.xls is listed as malicious by 40/62 Vendors. Threat categories are trojan/downloader and the Family labels of x97, mdloadr and tioibesj.

40 / 62
Community Score

40/62 security vendors flagged this file as malicious

1df68d55968bb6d2db4d0d18155188a03a42850f543c8595166ac6987df820
research-1646684671.xls
xls calls wmi

Size
648.50 KB

Last Analysis Date
7 days ago

Reanalyze Similar More

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 24

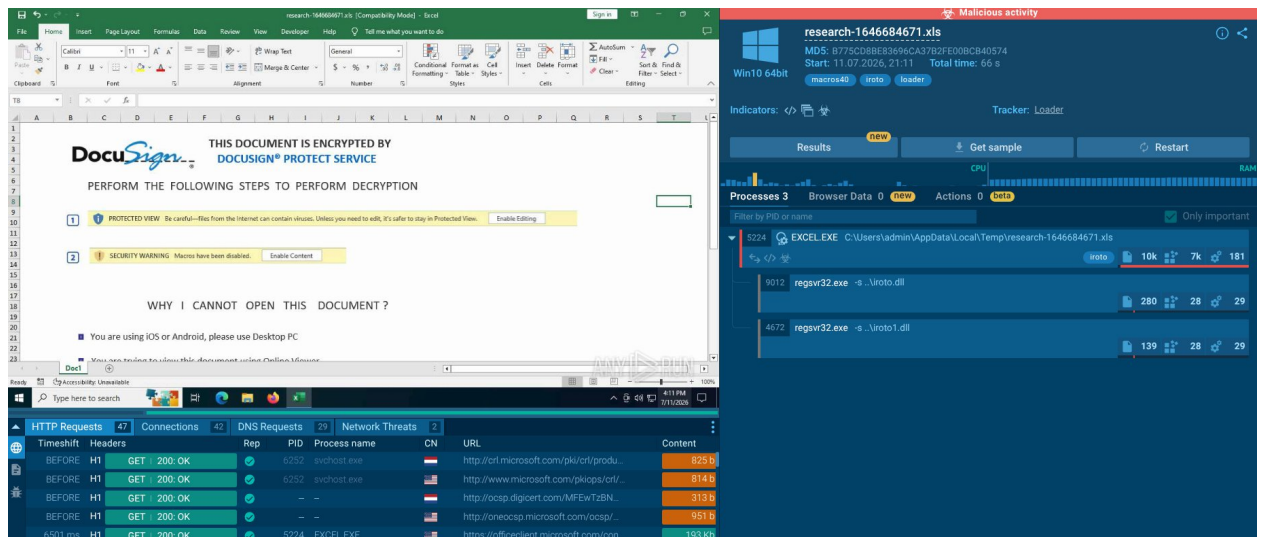
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.x97/mdloadr Threat categories trojan downloader Family labels x97 mdloadr tioibesj

Security vendors' analysis Do you want to automate checks?

AliCloud	Trojan.MSOffice/Downloader.amd	ALYac	Trojan.GenericKD.46481356
Antiy-AVL	Trojan(Downloader)/Macro.Agent	Arcabit	Trojan.Generic.D2C53FCC
Avast	VBS/Malware-gen	AVG	VBS/Malware-gen
Avira (no cloud)	X97/Agent.B2	BitDefender	Trojan.GenericKD.46481356
CTX	Xls.trojan.x97m	Cyren	Malicious (score: 99)
DrWeb	X97M.DownLoader.782	Emsisoft	Trojan.GenericKD.46481356 (8)

The xls file can also be found on AnyRun where it is indicated as performing malicious activities.

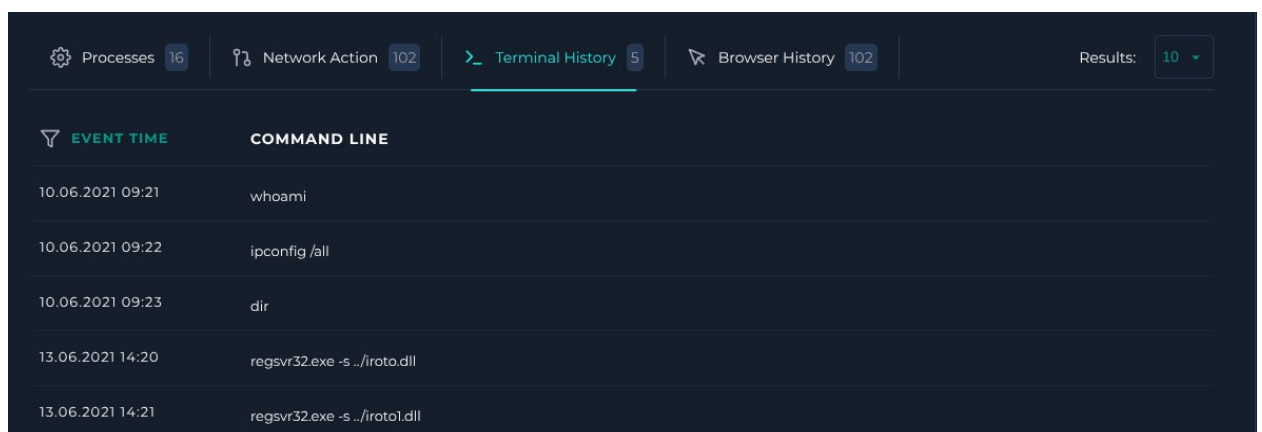


Once the macros run, it is apparent in the AnyRun process tree that EXCEL.EXE immediately spawns a child process called regsvr32.exe.

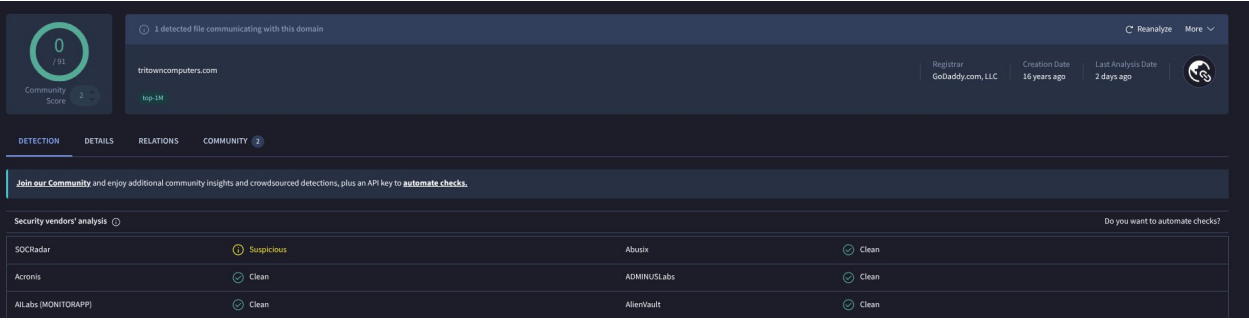
- regsvr32.exe (Microsoft Register Server) is a legitimate Windows command-line utility used to register and unregister Object Linking and Embedding (OLE) controls like DLLs.
- Threat actors abuse it to load malicious DLL files, bypassing application whitelisting.

The macro tells regsvr32.exe to run the two DLL files (iroto.dll and iroto1.dll). The -s parameter in the command line (regsvr32.exe -s ..\iroto.dll) tells the tool to run "silently," meaning no pop-up boxes will alert the user that a Trojan is actively installing in the background.

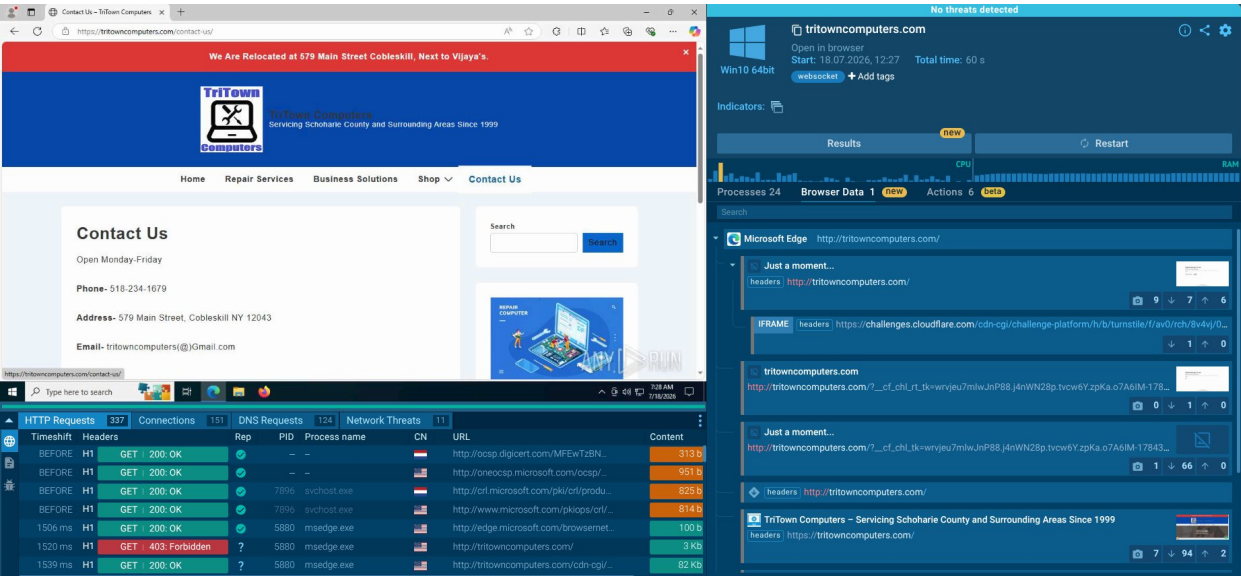
On conducting endpoint analysis it can be confirmed that the user accessed the file and that it executed on the system.



When performing OSINT on the sender email domain everything comes up clean.



AnyRun analysis suggests them to be a legitimate business.



This would suggest that they may have been compromised and are currently being exploited as a launchpad for malicious activities.

AbuseIPDB analysis suggests that this could be a front. They have been reported 34 times.

Reporter	IoA Timestamp (UTC)	Comment	Categories
cheatmaster.store	2025-08-10 16:45:56 (11 months ago)	Detected as open proxy - identified as honeypot	Hacking
vexhost.pl	2025-06-07 00:00:00 (1 year ago)	[VOPI] Open proxy, detected source: PS.	Open Proxy
vexhost.pl	2025-06-07 00:00:00 (1 year ago)	[VOPI] Open proxy, detected source: PS.	Open Proxy
vexhost.pl	2025-06-07 00:00:00 (1 year ago)	[VOPI] Open proxy, detected source: GitHub.	Open Proxy
vexhost.pl	2025-06-07 00:00:00 (1 year ago)	[VOPI] Open proxy, detected source: GitHub.	Open Proxy
Anonymous	2024-09-08 13:05:47 (1 year ago)	Scam trading site	Fraud Orders Phishing
ISPLtd	2024-09-05 06:25:42 (1 year ago)	Sep 5 03:25:40 SRC=66.235.200.146 PROTO=TCP SP T=80 DPT=29920 SYN Sep 5 03:25:41 SRC=66.235.20 0.146 ... show more	Port Scan
KPS	2024-08-21 16:25:57 (1 year ago)	PortscanM	Port Scan
AUS Honeypots	2024-08-21 15:38:09 (1 year ago)	Honeypot hits: 16 hits in 127 second(s)	Port Scan
Bad Ass Destroyer	2024-05-18 21:45:32 (2 years ago)	Impersonates https://www.bitpay.co.il/ the organization.	Phishing
scambhunter	2024-04-29 21:18:00 (2 years ago)	Scammers !	Fraud Orders Phishing Web Spam Spoofing Web App Attack

Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a True Positive phishing attack targeting an internal user. The threat actor utilised a likely compromised legitimate business domain (tritowncomputers[.]com) to bypass email reputation filters and deliver a ZIP archive. The archive contained a macro-enabled Excel document that successfully executed two malicious IROTO Trojan DLLs on the endpoint using the native Windows regsvr32.exe utility.

Incident Details

- **Alert Name:** SOC146 - Phishing Mail Detected - Excel 4.0 Macros
- **Target Account:** lars[@]letsdefend[.]io
- **Sender Address:** trenton[@]tritowncomputers[.]com
- **Sender IP:** 24[.]213.228.54
- **Email Subject:** RE: Meeting Notes

Analysis and Evidence

- **Delivery & Evasion:** The email contained a ZIP attachment (11f44531fb088d31307d87b01e8eabff) secured with the password "infected" to evade secure email gateways.
- **The Loader:** The extracted file, research-1646684671.xls, utilises a fraudulent DocuSign template to trick the user into enabling Excel 4.0 macros.
- **Execution:** Terminal history confirms the user interacted with the payload. The macros spawned regsvr32.exe to silently load and execute the hidden payloads (regsvr32.exe -s ..\iroto.dll and regsvr32.exe -s ..\iroto1.dll).

Threat Intelligence

- **File Analysis:** The extracted files are absent from the internal CTI database but are heavily flagged on VirusTotal. The Excel loader is flagged by 40/62 vendors as a Trojan/Downloader. The payloads, iroto.dll and iroto1.dll, are flagged by 18/62 and 21/68 vendors respectively, confirming they belong to the IROTO/abtrojan family.
- **Domain OSINT:** While VirusTotal shows 0/91 detections for the sender domain tritowncomputers[.]com, AnyRun analysis shows a seemingly legitimate local computer repair business. However, AbuseIPDB reports indicate the associated infrastructure has been reported 34 times for hosting open proxies, port scanning, and phishing, indicating a potentially compromised environment.

Actions Taken

1. Analysed email logs and safely extracted the ZIP archive to obtain file hashes.
2. Conducted OSINT on the sender domain, confirming it is likely a compromised legitimate business used to bypass reputation filters.
3. Mapped the execution flow in AnyRun, confirming the use of fake DocuSign lures and regsvr32.exe to silently load malicious DLLs.
4. Reviewed endpoint terminal history, confirming the payloads successfully executed on the host machine.
5. Isolated the endpoint from the network.
6. Declared the alert a True Positive and escalated.

Let's Defend Result

SOC146 - Phishing Mail Detected - Excel 4.0 Macros

True Positive (+5 Point)

Check If Someone Opened the Malicious File/URL? (+5 Point)

Check If Mail Delivered to User? (+5 Point)

Analyze Url/Attachment (+5 Point)

Are there attachments or URLs in the email? (+5 Point)